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Assignment No: 9

Problem Statement:

Use appropriate algorithm and perform sentiment analysis for Indian Language.

Objective:

- To understand the challenges of sentiment analysis in Indian languages such as Hindi, Marathi, and Tamil.
- To load and utilize a pre-trained multilingual XLM-RoBERTa model for Indian language sentiment classification.
- To tokenize Indian language text using a multilingual SentencePiece-based tokenizer.
- To classify sentences as Positive, Neutral, or Negative with associated confidence scores.
- To evaluate the model's performance across multiple Indian language scripts including Devanagari and Tamil.

S/W Packages and H/W apparatus used:

- Operating System: Windows/Linux/MacOS
- Kernel: Python 3.x
- Tools: Google Colab / Jupyter Notebook
- Hardware: CPU with minimum 4GB RAM; GPU recommended for faster inference
- Libraries and packages used: Transformers (Hugging Face), PyTorch, SentencePiece, torch.nn.functional

Theory:

Sentiment analysis is the NLP task of determining the emotional polarity (positive, negative, or neutral) expressed in a piece of text. While extensively studied for English, Indian languages present unique challenges due to their diverse scripts, morphological complexity, code-mixing, and limited annotated resources.

Challenges of Indian Language Sentiment Analysis:

Indian languages such as Hindi, Marathi, Tamil, Telugu, and Bengali use distinct scripts (Devanagari, Tamil, Telugu, etc.) with rich morphology and agglutinative word structures. Sentiment lexicons and annotated corpora for Indian languages are far less extensive than for English. Additionally, social media text in India frequently exhibits code-mixing (Hinglish, Tanglish) and script switching, making rule-based approaches largely ineffective.

XLM-RoBERTa (Cross-lingual Language Model - Robustly Optimized BERT):

XLM-RoBERTa is a multilingual variant of RoBERTa trained by Meta AI on 2.5 terabytes of filtered CommonCrawl data covering 100 languages, including major Indian languages. It uses a shared multilingual SentencePiece vocabulary of 250,000 tokens, enabling it to encode text from any of the supported languages in a shared semantic space. This cross-lingual pretraining allows the model to transfer sentiment knowledge learned from resource-rich languages to low-resource Indian languages.

cardiffnlp/twitter-xlm-roberta-base-sentiment:

This model is XLM-RoBERTa-base fine-tuned by Cardiff NLP on multilingual Twitter data for three-class sentiment classification: Negative (0), Neutral (1), and Positive (2). Twitter data naturally contains informal, short-form text across many languages, making this model well-suited for colloquial Indian language sentiment analysis. The model outputs logits for three classes, which are converted to probabilities using the softmax function.

Softmax and Confidence Scores:

The model's raw output (logits) is passed through torch.nn.functional.softmax() to produce a probability distribution over the three sentiment classes. The class with the highest probability is selected as the predicted sentiment using torch.argmax(). The corresponding probability value is reported as the confidence score, providing an interpretable measure of the model's certainty.

Methodology:

- Install required libraries: transformers, torch, and sentencepiece.
- Import AutoTokenizer and AutoModelForSequenceClassification from Hugging Face Transformers, and torch.nn.functional for softmax.
- Load the pre-trained cardiffnlp/twitter-xlm-roberta-base-sentiment model and its multilingual SentencePiece tokenizer from the Hugging Face model hub.
- Define a list of five Indian language sentences: two in Hindi, two in Marathi (Devanagari script), and two in Tamil script, covering both positive and negative sentiments.
- For each sentence, tokenize the input using the loaded tokenizer with return_tensors='pt', truncation=True, and padding=True.
- Pass the tokenized inputs through the model to obtain logits, then apply softmax to convert logits to class probabilities.
- Use torch.argmax() to identify the predicted sentiment class index and map it to the label list ['Negative', 'Neutral', 'Positive'].
- Print the original sentence, the predicted sentiment label, and the confidence score rounded to 4 decimal places for each input.

Use Cases:

- Use Case 1 - Product Review Analysis for Indian E-commerce: Classifying customer reviews written in Hindi, Marathi, or Tamil as positive or negative helps platforms like Flipkart and Meesho aggregate regional sentiment signals for product quality monitoring without requiring separate language-specific models.
- Use Case 2 - Social Media Brand Monitoring: Tracking brand mentions and public sentiment on Indian-language Twitter/X posts using a multilingual model enables companies to monitor reputation across Hindi, Tamil, Marathi, and other regional language communities in real time.
- Use Case 3 - Government Public Feedback Analysis: Analyzing citizen feedback submitted in regional Indian languages on government portals and helpline platforms helps authorities gauge public sentiment about policies, schemes, and services without language barriers.

Advantages:

- A single multilingual XLM-RoBERTa model handles Hindi, Marathi, Tamil, and 97 other languages without requiring separate language-specific models.
- Fine-tuning on multilingual Twitter data makes the model robust to informal, short-form, and colloquial Indian language text.
- Confidence scores from softmax provide interpretable uncertainty estimates alongside predictions.
- The Hugging Face pipeline requires minimal code -model loading, tokenization, inference, and decoding are handled in a few lines.
- No Indian language preprocessing (tokenization, stop word removal) is required; the SentencePiece tokenizer handles all scripts natively.

Limitations:

- The model was fine-tuned on Twitter data; it may underperform on formal Indian language text such as news articles or government documents.
- Low-resource Indian languages (e.g., Odia, Assamese, Konkani) are underrepresented in the training corpus, leading to weaker performance.
- Code-mixed text (Hinglish, Tanglish) with Roman script representations of Indian words may not be tokenized optimally.
- The model provides three-class sentiment only (Positive/Neutral/Negative) and cannot detect more nuanced emotions like anger, fear, or surprise.
- CPU-based inference on the 1.1GB XLM-RoBERTa model is slow; production deployments require GPU acceleration or model distillation.

Applications:

- Regional language social media monitoring: Real-time sentiment tracking of Hindi, Marathi, Tamil, and other Indian language posts for brand and political analysis.
- Customer support automation: Classifying incoming support tickets written in Indian languages by sentiment to prioritize escalations.
- Opinion mining for Indian elections: Analysing multilingual political campaign content and voter responses on social platforms.
- News sentiment aggregation: Tracking public mood from Indian-language news headlines and reader comments across regional publications.
- Healthcare feedback systems: Classifying patient satisfaction feedback submitted in regional Indian languages at hospitals and telemedicine platforms.

Conclusion:

This assignment performed sentiment analysis on Indian language text using the cardiffnlp/twitter-xlm-roberta-base-sentiment model, a multilingual XLM-RoBERTa model fine-tuned on multilingual Twitter data. Five sentences in Hindi, Marathi, and Tamil were classified into Positive, Neutral, or Negative sentiment categories with high confidence scores (0.80 to 0.93). The model correctly identified positive sentiments in Hindi and Marathi praise sentences, negative sentiments in Marathi and Tamil criticism sentences, and handled all three distinct scripts -Devanagari and Tamil -without any language-specific preprocessing. The experiment demonstrates that modern cross-lingual transformer models can effectively bridge the resource gap for Indian language sentiment analysis, enabling practical deployment across India's linguistically diverse digital landscape.